A Deep Learning Approach for Brain Tumor Classification Using MRI Images

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Guide: **Dr. P. Sreekanth** Team DS-A-T6 · Review 2, PROJ2999 Capstone Project Introduction

Introduction

Early, accurate typing of a brain tumor shapes its treatment. MRI is the main imaging tool, but reading slices by hand is slow, needs scarce expertise and varies between readers.

Deep CNNs report 97–99 % accuracy on public MRI benchmarks [1], [2], yet nearly every figure comes from one dataset and is given as one overall number.

Problem statement: for four-class MRI classification (glioma, meningioma, pituitary, no tumor), how well do such models hold up on a second dataset, which class fails most, and at what cost? This project answers all three under one documented, reproducible protocol.

Literature Review

13 papers reviewed (2021–2026). Representative results:

Ref.	Approach	Reported	Limitation
[1]	EfficientNet + ResNet-50 feature fusion	98.95 % Figshare	one dataset only
[2]	SE-ResNet101 + ensemble classifier	99.67 % Figshare	costly, hard to interpret
[3]	Eight CNNs + ViT, Grad-CAM, LIME	98.17 % 7,023 images	some models attend to irrelevant regions
[4]	VGG16, InceptionV3, MobileNetV2	86.00 % 253 images	small data: accuracy drops

Research Gaps

G1 Accuracy proven on one benchmark only: 98.95 % [1] and 99.67 % [2] on Figshare, vs 86 % on 253 other images [4].

G2 A single overall accuracy hides weak classes and false positives.

G3 No common evaluation protocol; model cost and interpretability are rarely reported [2], [3].

Project Objectives

O1 Build an EfficientNet-B0 transfer-learning classifier [5], benchmarked against ResNet-50 and a scratch CNN, reaching **macro-F1 ≥ 0.95** on the 7,023-image test set with **< 10 M parameters** (G3).

O2 Evaluate it on a second dataset (Figshare T1-CE: 3,064 slices, 233 patients, patient-level splits); report per-class precision, recall, F1, specificity and Grad-CAM maps, with **recall ≥ 0.90 for every class** (G1, G2).

Proposed Methodology

1. **Input** — 7,023 images (4 classes) + 3,064 slices
2. **Curation** — pHash de-duplication, frozen splits
3. **Preprocess** — brain crop, 224 × 224, augmentation
4. **Training** — EfficientNet-B0 vs ResNet-50 vs CNN
5. **Evaluation** — per-class P / R / F1, ROC, cost
6. **Output** — class, confidence, Grad-CAM heatmap

Fig. 1: Proposed methodology, input to explained output

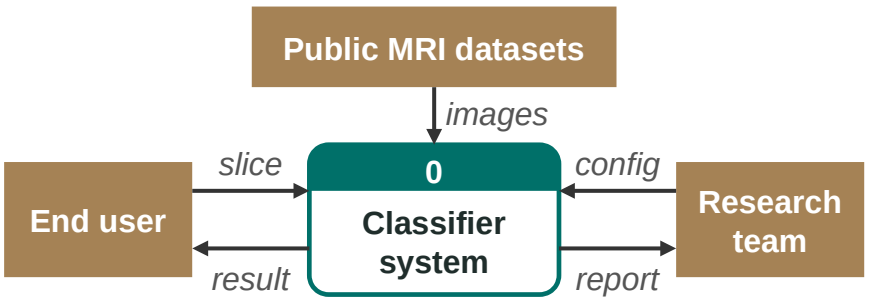


Fig. 2: Level-0 DFD (Gane–Sarson notation)

Tools/ Technology / Algorithms

Tools: Python, PyTorch + torchvision, scikit-learn, OpenCV, Albumentations, pytorch-grad-cam, Streamlit, GitHub.

Technology: ImageNet transfer learning; Grad-CAM explainability; Colab / Kaggle T4 GPU.

Algorithms: EfficientNet-B0 [5] (5.3 M params, primary), ResNet-50 (25.6 M, baseline), scratch CNN (ablation); AdamW, cosine LR, class-weighted cross-entropy.

References

[1] M. Aamir et al., "A deep learning approach for brain tumor classification using MRI images," *Comput. Electr. Eng.*, vol. 101, 2022, Art. no. 108105.

[2] M. Aamir et al., "An automated deep learning framework for brain tumor classification using MRI imagery," *Sci. Rep.*, vol. 15, 2025, Art. no. 17593.

[3] E. F. Gomes and R. S. Barbosa, "Deep learning approaches for brain tumor classification in MRI scans: An analysis of model interpretability," *Appl. Sci.*, vol. 16, no. 2, 2026, Art. no. 831.

[4] Z. N. I. Zailan et al., "Deep learning approach for prediction of brain tumor from small number of MRI images," *JOIV*, vol. 6, no. 2-2, pp. 581–586, 2022.

[5] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.